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Exercises are from Stephen Abbott's *Understanding Analysis*, Second Edition.

Problem 1: Weierstrass M-Test

Exercise 6.4.1

Theorem 1: For each $n \in \mathbb{N}$, let f_n be a function defined on a set $A \subseteq \mathbb{R}$, and let $M_n > 0$ be a real number satisfyin

$$|f_n(x)| \leq M_n$$

for all $x \in A$. If $\sum_{n=1}^{\infty} M_n$ converges, then $\sum_{n=1}^{\infty} f_n$ converges uniformly on A .

Proof: Let $\varepsilon > 0$ be given. By the Cauchy Criterion for Series applied to $\sum_{n=1}^{\infty} M_n$, there exists some $N \in \mathbb{N}$ such that whenever $n > m \geq N$ we have

$$|M_{m+1} + M_{m+2} + \dots + M_n| < \varepsilon.$$

Since every $M_i > 0$, this is equivalent to

$$M_{m+1} + M_{m+2} + \dots + M_n < \varepsilon.$$

Then for any $x \in A$ we have

$$\begin{aligned} |f_{m+1}(x) + f_{m+2}(x) + \dots + f_n(x)| \\ &\leq |f_{m+1}(x)| + |f_{m+2}(x)| + \dots + |f_n(x)| \\ &\leq M_{m+1} + M_{m+2} + \dots + M_n \\ &< \varepsilon. \end{aligned}$$

By the Cauchy Criterion for Uniform Convergence of Series, since the choice of N was independent of x , $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on A . ■

Problem 2: A trick for proving continuity at a point

Exercise 6.4.5

Proof idea

There is a cool idea happening in part (b). We can't apply the Weierstrass M-Test on all of $(-1, 1)$ at once, because there's no single M_n that will work for all x in this interval: the defining series converges everywhere in this interval but its sum can be made unboundedly large by getting closer and closer to 1. Another way to put it is that there's no single convergent series of M_n that dominates $\left|\frac{x^n}{n}\right|$ for *all* $x \in (-1, 1)$. For any candidate M_n , there exist x close enough to 1 such that $|x^n/n| > M_n$. If the M_n get big enough to dominate, then their sum will diverge.

But the trick is that for any specific $x_0 \in (-1, 1)$ we can construct an interval that contains it, and produce an M_n that bounds it there.

Note that we can't just use singleton sets $\{x_0\}$ and define f on those to get around this problem, because continuity on singleton sets is trivial, or, to be more precise, doesn't extend to continuity on an open set containing the point. What we really want is to say that the function f defined on $(-1, 1)$ is continuous for any fixed $x_0 \in (-1, 1)$. If we tried to do this with singleton sets, we would have proved that $f : \{x_0\} \rightarrow \mathbb{R}$ is continuous at x_0 . But this can't be extended to the statement that $f : (-1, 1) \rightarrow \mathbb{R}$ is continuous at x_0 . What's good about the idea that works is that we prove that $f : [-a, a] \rightarrow \mathbb{R}$ is continuous at $x_0 \in [-a, a]$, from which we can say that $f : (-1, 1) \rightarrow \mathbb{R}$ is continuous at $x_0 \in (-1, 1)$.

Proofs

(a)

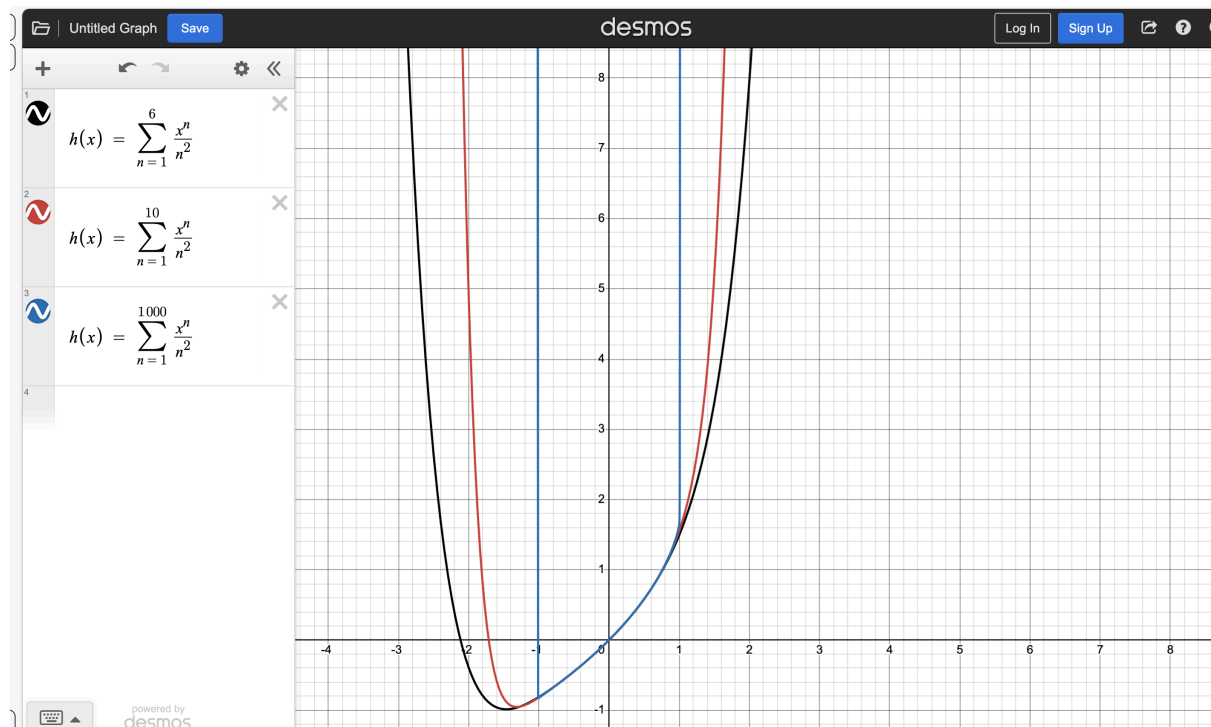


Figure 1: Graphs of $h(x)$ with more and more terms

Theorem 2: The function h defined by the series

$$h(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2} = x + \frac{x^2}{4} + \frac{x^3}{9} + \dots$$

is continuous on $[-1, 1]$.

Proof: For $x \in [-1, 1]$,

$$|x^n| = |x|^n \leq 1.$$

Let $f_n(x) = \frac{x^n}{n^2}$. Then

$$|f_n(x)| \leq \frac{1}{n^2}$$

when $x \in [-1, 1]$.

Let $M_n = \frac{1}{n^2}$. Then by the Cauchy Condensation Test (or, as proved several homeworks ago, since it is the series $\sum_{n=1}^{\infty} 1/n^p$ with $p = 2 > 1$), the series $\sum_{n=1}^{\infty} M_n$ converges. Therefore by the Weierstrass M-Test, the series $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly for $x \in [-1, 1]$. Therefore, since each f_n is continuous (because polynomial), by the Term-by-term Continuity Theorem, $h(x)$ is continuous on $[-1, 1]$. ■

(b)

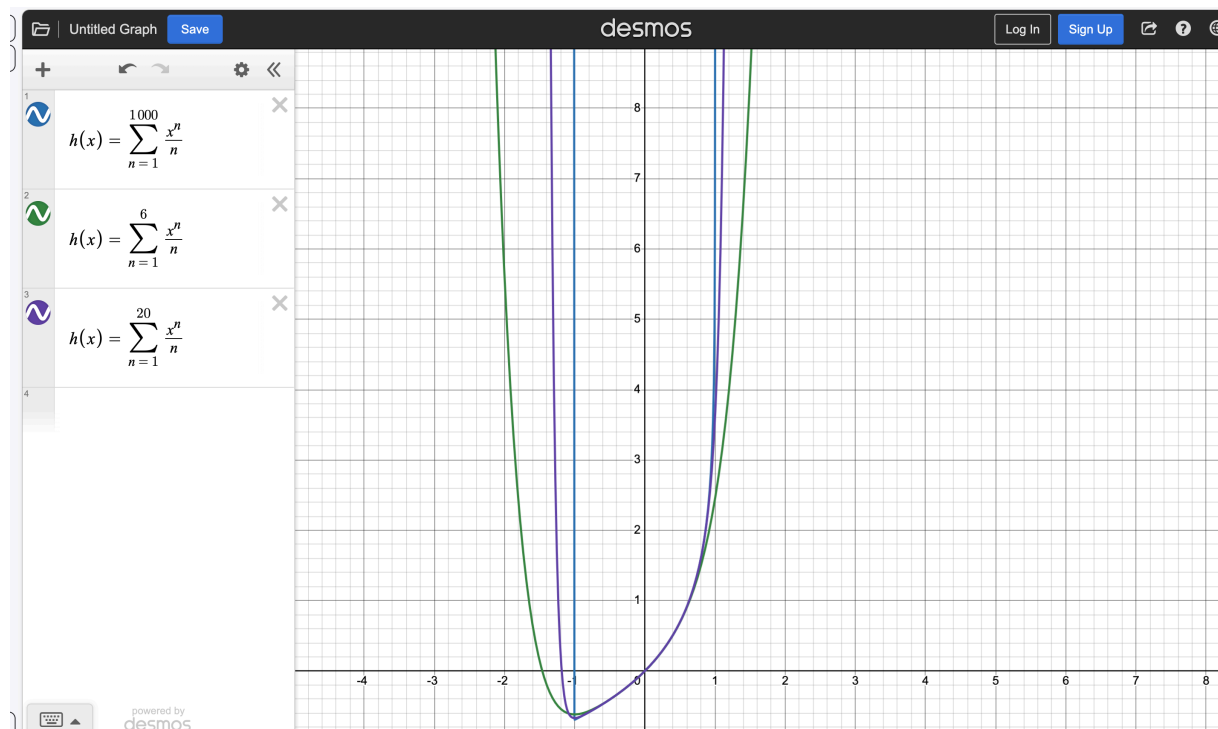


Figure 2: Graphs of $f(x)$ with more and more terms

Theorem 3: Let

$$f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n} = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$$

Then for any fixed $x_0 \in (-1, 1)$, f is continuous at x_0 . Thus, f is continuous on $(-1, 1)$.

Proof: Since $x_0 \in (-1, 1)$, we have $|x_0| < 1$. So there exists some $a \in (0, 1)$ such that $|x_0| < a < 1$. (Note to myself: this isn't a consequence of the completeness of the reals. Just from being an ordered field. $\mathbb{Q}$ would have such an a as well.)

Let $M_n = a^n/n$. Let f_n be defined on the set $A = [-a, a]$ given by

$$f_n(x) = \frac{x^n}{n}.$$

For every $x \in A$ we have

$$|x| \leq a$$

so

$$|x^n| = |x|^n \leq a^n$$

so

$$\left| \frac{x^n}{n} \right| \leq \frac{a^n}{n}$$

so

$$|f_n(x)| \leq M_n.$$

We compute

$$\begin{aligned} \lim \left| \frac{M_{n+1}}{M_n} \right| &= \lim \left| \frac{a^{n+1}/(n+1)}{a^n/n} \right| \\ &= \lim |a| \left| \frac{n}{n+1} \right| \\ &= \lim |a| \left| \frac{1}{1+1/n} \right| \\ &= |a| \\ &< 1 \end{aligned}$$

by the Algebraic Limit Theorem. So by the Ratio Test, the series $\sum_{n=1}^{\infty} M_n$ converges.

So by the Weierstrass M-Test, $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly for all $x \in A$. Therefore, since each f_n is continuous on A since it is a polynomial, f is continuous on A . So $f : [-a, a] \rightarrow \mathbb{R}$ is continuous at x_0 .

But then $f : (-1, 1) \rightarrow \mathbb{R}$ is also continuous at x_0 , since every sequence $(x_n) \subseteq (-1, 1)$ that converges to x_0 is eventually entirely in $[-a, a] \subseteq (-1, 1)$, so $(f(x_n)) \rightarrow f(x_0)$.

So f is continuous on $(-1, 1)$. ■

Problem 3: Uniform convergence of an alternating harmonic-like series

Exercise 6.4.6

Proof idea

I can see two ways of doing this but both involve a little bit of a “trick”. One is to combine adjacent terms and apply the Weierstrass M-Test. The other is to use a fact we proved on the way to proving the Alternating Series Test via the Cauchy Criterion.

Proofs

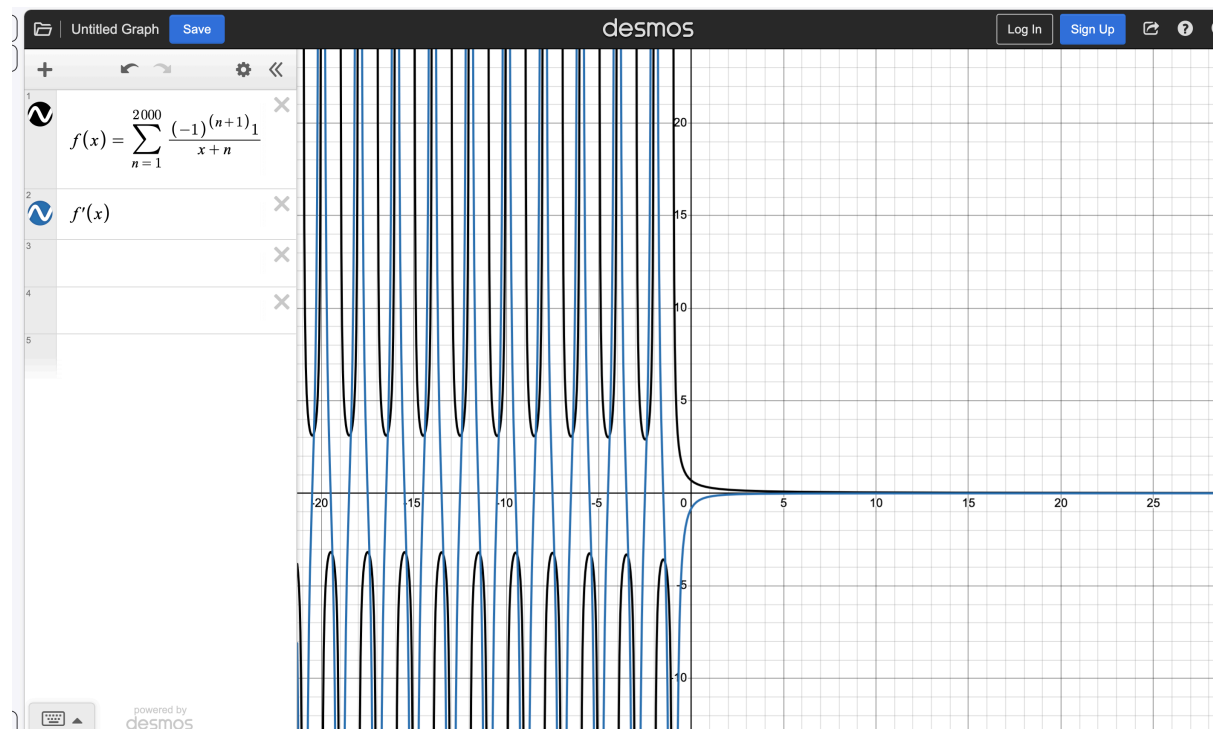


Figure 3: Graphs of $f(x)$ and the series obtained by differentiating term-by-term. Note the crazy behavior when $x < 0$

Let

$$f(x) = \frac{1}{x} - \frac{1}{x+1} + \frac{1}{x+2} - \frac{1}{x+3} + \dots$$

and

$$f_n(x) = \frac{1}{x+n-1}$$

and

$$g_n(x) = (-1)^{n+1} f_n(x).$$

Lemma 1: f is defined for all $x > 0$.

Proof: We use the Alternating Series Test. For all $x > 0$,

$$\frac{1}{x} > \frac{1}{x+2} > \dots > \frac{1}{x+n} > \frac{1}{x+n+1} > \dots$$

and

$$\frac{1}{x+n} = \frac{1/n}{x/n+1} \rightarrow \frac{0}{1} = 0$$

by the Algebraic Limit Theorem.

Therefore the series defining f converges when $x > 0$ so f is defined. ■

I will restate here (without proof) a fact I proved on the way to the Alternating Series Test. I did this with a bunch of casework involving the parity of $m - n$ and so forth, and you pointed out an inductive argument. The fact is:

Lemma 2: Let (a_k) be a sequence satisfying

(i) $a_1 \geq a_2 \geq \dots \geq a_k \geq \dots$, and

(ii) $(a_k) \rightarrow 0$.

Let $n > m \geq 1$. Then

$$|a_{m+1} - a_{m+2} + a_{m+3} - \dots \pm a_n| \leq a_{m+1}.$$

Another way to state this is: let

$$s_n = a_1 - a_2 + a_3 - \dots \pm a_n,$$

the sequence of partial sums of the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n.$$

Then

$$|s_n - s_m| \leq a_{m+1}.$$

Lemma 3: The series

$$\sum_{n=1}^{\infty} g_n(x)$$

converges uniformly for $x > 0$.

Proof: Let $\varepsilon > 0$ be given. Pick any $N \in \mathbb{N}$ such that

$$\frac{1}{N} < \varepsilon.$$

Such an N exists because $(\frac{1}{n}) \rightarrow 0$ and by the Archimedean Property.

Let $n > m \geq N$. Then by the lemma above, for any $x > 0$ we have

$$\begin{aligned}
& |g_{m+1}(x) + g_{m+2}(x) + \dots + g_n(x)| \\
& \leq |g_{m+1}(x)| \\
& = f_{m+1}(x) \\
& = \frac{1}{x+m} \\
& < \frac{1}{m} \\
& \leq \frac{1}{N} \\
& < \varepsilon.
\end{aligned}$$

The result follows by the Cauchy Criterion for Uniform Convergence of Series. ■

Since each term in the series is continuous, by uniform convergence we have continuity of the function f on $(0, \infty)$.

Here's another proof of continuity using the Weierstrass M-Test and the trick from Problem 2:

Proof: Let (s_n) be the sequence of partial sums of the original series. By the Alternating Series Test argument above, this sequence converges.

By pairing adjacent terms we have

$$\begin{aligned}
& \frac{1}{x} - \frac{1}{x+1} + \frac{1}{x+2} - \frac{1}{x+3} + \dots \\
& = \frac{1}{x(x+1)} + \frac{1}{(x+2)(x+3)} + \dots
\end{aligned}$$

Let

$$h_n(x) = \frac{1}{(x+2n-2)(x+2n-1)}.$$

We claim that the original series equals

$$\sum_{n=1}^{\infty} h_n(x)$$

since the sequence of partial sums of this series is (s_{2n}) , which is a subsequence of (s_n) , and subsequences of convergent sequences converge to the same limit.

Let x be given and pick some a such that $0 < a < x$. Let

$$M_n = \begin{cases} 1/a(a+1) & \text{if } n = 1 \\ 1/n^2 & \text{if } n > 1 \end{cases}$$

.

Then for all $x > a$ and $n \geq 1$ we have

$$h_n(x) \leq M_n.$$

The series

$$\sum_{n=1}^{\infty} M_n = \frac{1}{a(a+1)} + \sum_{n=2}^{\infty} \frac{1}{n^2}$$

converges since it's a finite number plus the tail of a series known to converge.

So the series converges uniformly on (a, ∞) by the Weierstrass M-Test, so f is continuous on (a, ∞) . By the same argument as in Problem 2, $f : (0, \infty) \rightarrow \mathbb{R}$ is continuous at x , so f is continuous on $(0, \infty)$. ■

Lemma 4: f is differentiable on $(0, \infty)$.

Proof: Let $x \in (0, \infty)$ be given and pick some a such that $0 < a < x$.

Let

$$M_n = \frac{1}{(a + n - 1)^2}.$$

Then

$$\sum_{n=1}^{\infty} M_n = \frac{1}{a^2} + \sum_{n=2}^{\infty} M_n.$$

The second series above,

$$\sum_{n=2}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{(a + n)^2}$$

converges by the comparison test against $\sum_{n=1}^{\infty} 1/n^2$. Since $\sum_{n=1}^{\infty} M_n$ is a finite term plus this convergent series, it converges.

Then the absolute value of the derivative of each term of the series for f is

$$|g'_n(x)| = \left| -\frac{1}{(x + n - 1)^2} \right|$$

so

$$|g'_n(x)| \leq M_n$$

for all $x \in [a, \infty)$. Therefore, by the Weierstrass M-Test, the series of derivatives of terms converges uniformly on $[a, \infty)$.

Since f converges on $(0, \infty)$ by the Term-by-Term Differentiability Theorem we have that $f : [a, \infty) \rightarrow \mathbb{R}$ is differentiable at x . By a similar argument as in Problem 2, since differentiability is a local property, we have that $f : (0, \infty) \rightarrow \mathbb{R}$ is differentiable at x , so it's differentiable on all of $(0, \infty)$.

To make that "local property" bit rigorous: since $f : [a, \infty) \rightarrow \mathbb{R}$ is differentiable at x , the limit

$$\lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x} = L$$

exists. Let $g(y)$ denote the difference quotient above as a function of y . Every sequence $(y_n) \subseteq (0, \infty)$ converging to x is eventually in $[a, \infty)$ (by the definition of convergence and the triangle inequality). So $(g(y_n)) \rightarrow L$ when g is considered as a function defined on $(0, \infty) \setminus \{x\}$, therefore $f : (0, \infty) \rightarrow \mathbb{R}$ is differentiable at x . ■

Problem 4: Continuity and differentiability of another series

Exercise 6.4.9

Proof idea

This is more practice with the Weierstrass M-Test. I found another place to use the AM-GM inequality, but we could have just done it in the usual way by taking derivatives and setting them to zero and looking at signs. Either way, proving uniform convergence of the derivative series gives us differentiability and *continuous* differentiability in one fell swoop.

Proofs

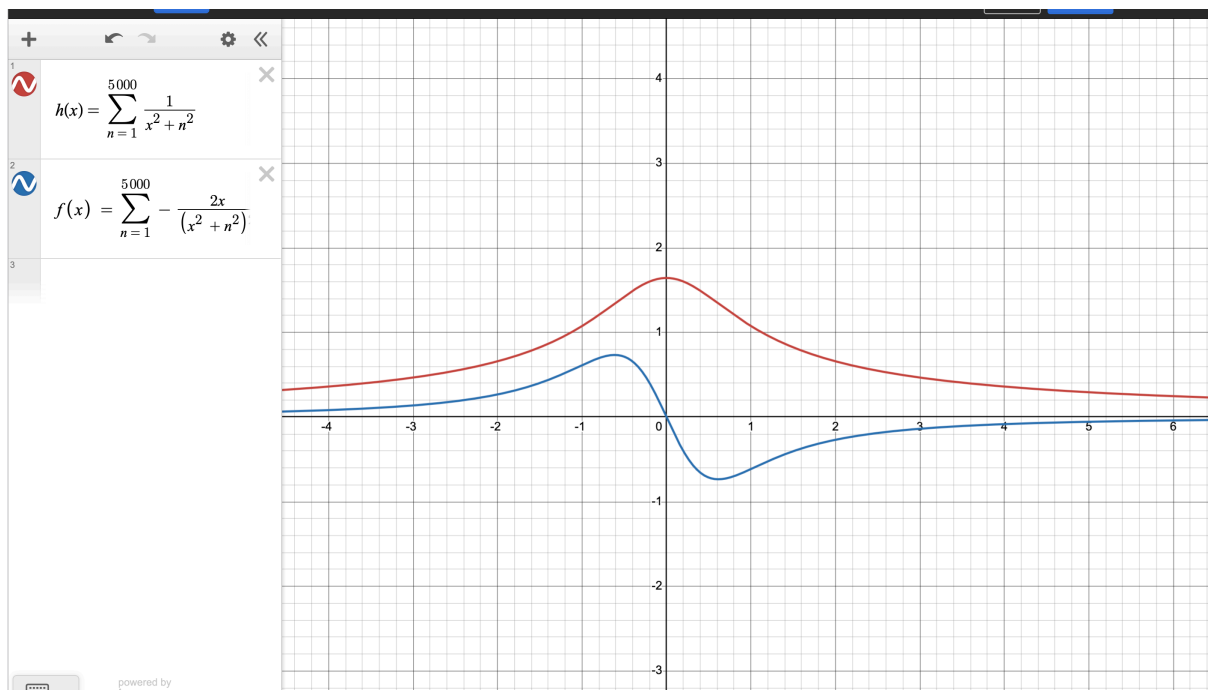


Figure 4: Graphs of $h(x)$ and the series obtained by differentiating term-by-term

Let

$$h(x) = \sum_{n=1}^{\infty} \frac{1}{x^2 + n^2}$$

and

$$h_n(x) = \frac{1}{x^2 + n^2}.$$

Theorem 4: h is a continuous function defined on all of $\mathbb{R}$.

Proof: Let $M_n = 1/n^2$. Then

$$|h_n(x)| \leq M_n$$

since $x^2 \geq 0$. By the Weierstrass M-Test, the series $\sum_{n=1}^{\infty} h_n(x)$ converges uniformly on all of $\mathbb{R}$. Since each h_n is continuous on all of $\mathbb{R}$, so is h . ■

Theorem 5: h is continuously differentiable on all of $\mathbb{R}$.

Proof: By the AM-GM inequality applied to x^2 and n^2 , we have

$$\frac{x^2 + n^2}{2} \geq n|x|$$

so

$$\frac{2|x|}{x^2 + n^2} \leq \frac{1}{n}.$$

Also,

$$\frac{1}{x^2 + n^2} \leq \frac{1}{n^2},$$

so

$$|h'_n(x)| = \left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq \frac{1}{n^3}.$$

The series $\sum_{n=1}^{\infty} 1/n^3$ converges since it's of the form $\sum_{n=1}^{\infty} 1/n^s$ with $s = 3 > 1$. So by the Weierstrass M-Test, $\sum_{n=1}^{\infty} h'_n(x)$ converges uniformly on all of $\mathbb{R}$. Since the original series converges, h is differentiable on all of $\mathbb{R}$, and its derivative is given by the series

$$h'(x) = \sum_{n=1}^{\infty} h'_n(x).$$

Each term $h'_n(x)$ is continuous on all of $\mathbb{R}$ since it's a rational function whose denominator is always at least 1. Therefore, since the series for the derivative converges uniformly on all of $\mathbb{R}$, the derivative is continuous on all of $\mathbb{R}$. ■

Problem 5: A monotone function continuous at every irrational point

Exercise 6.4.10

Proof idea

The main thing is to prove that each u_n is continuous at every irrational point. For this I pulled out a little lemma that says, if a function is constant on some neighborhood U of c then the function is continuous at c .

We still need uniform convergence of the u_n at every $x \in \mathbb{R}$. The Weierstrass M-Test and geometric series do the job once again.

Proofs

Lemma 5: Let $f : A \rightarrow \mathbb{R}$ and $U \subseteq A$ be open. If f is constant on U then f is continuous on U .

Proof: Let $\varepsilon > 0$ be given and let c be any point in U . Since U is open, there exists some $\delta > 0$ such that the open ball $V_\delta(c) \subseteq U$. Then whenever $|x - c| < \delta$ we have $|f(x) - f(c)| = |f(c) - f(c)| = 0 < \varepsilon$ since f is constant on U therefore it's constant in $V_\delta(c)$ as well. So δ is a suitable reply to ε in the definition of continuity. ■

Another way to state this is (based on a comment by Nic):

Theorem 6: A function $f : X \rightarrow Y$ is continuous iff for some open cover of X , f is continuous on each open set in the cover.

Proof: ($\Rightarrow$)

Let $\varphi = \{O_\alpha : \alpha \in \alpha\}$ be an open cover of X such that f is continuous on every O_α . Let x be any point in X . Then $x \in O_b$ for some $O_b \in \varphi$ since φ is a cover of X . Let $V \subseteq Y$ be an open set containing $f(x)$. Since f is continuous on O_b , there exists an open set $U \subseteq O_b$ containing x such that $f(U) \subseteq V$. But then U is an open subset of X that contains x and whose image is contained in V , proving that $f : X \rightarrow Y$ is continuous at x .

($\Leftarrow$)

The open cover consisting of the whole space works: $\varphi = \{X\}$.

Or, here's another argument:

Let f be continuous on X and let $x \in X$ be any point. Let $V_x \subseteq Y$ be any open set containing $f(x)$. By continuity, there exists some open $U_x \subseteq X$ such that $x \in U_x$ and $f(U_x) \subseteq V_x$. Then

$$\varphi = \{U_x : x \in X\}$$

is an open cover of X . We claim that f is continuous on each U_x : let $y \in U_x$ be any point, let V_y be any open set containing $f(y)$. By continuity of f on X , there exists some open $U_y \subseteq X$ such that $y \in U_y$ and $f(U_y) \subseteq V_y$. Then $U_y \cap U_x$ is an open subset of U_x that contains y and whose image is contained in V_y . ■

Let $\{r_1, r_2, \dots\}$ be an enumeration of $\mathbb{Q}$. For each $r_n \in \mathbb{Q}$, define

$$u_n(x) = \begin{cases} 1/2^n & \text{for } x > r_n \\ 0 & \text{for } x \leq r_n. \end{cases}$$

Now let $h(x) = \sum_{n=1}^{\infty} u_n(x)$.

Lemma 6: Every $u_n(x)$ is continuous at every irrational point.

Proof: Let $y \in \mathbb{R}$ be irrational. Then either $y > r_n$ or $y < r_n$ and in both cases, u_n is constant in an open ball of radius $|y - r_n| > 0$ centered at y . By the lemma above, u_n is continuous at y . ■

Theorem 7: h is defined on all of $\mathbb{R}$ and continuous at every irrational point.

Proof: Let $M_n = 1/2^n$. Then $\sum_{n=1}^{\infty} M_n$ is a convergent geometric series.

Now, $u_n(x) \leq M_n$ for every $n \in \mathbb{N}$ and every $x \in \mathbb{R}$ because it is either zero or equal to M_n . Therefore $\sum_{n=1}^{\infty} u_n(x)$ converges uniformly for all of $\mathbb{R}$. So h is defined on all of $\mathbb{R}$.

Since each u_n is continuous at every irrational point, each partial sum is also continuous at every irrational point by the Algebraic Continuity Theorem. Therefore by the Continuous Limit Theorem applied to the partial sums, h is continuous at every irrational point. ■

Theorem 8: $h : \mathbb{R} \rightarrow \mathbb{R}$ is monotone.

Proof: Let $x \geq y$. For every r_n such that $y > r_n$, we have $x > r_n$ as well. So every nonzero term in the sum

$$\sum_{n=1}^{\infty} u_n(y)$$

is nonzero in

$$\sum_{n=1}^{\infty} u_n(x)$$

as well. Since every $u_n(x) \geq 0$, we have $h(x) \geq h(y)$. ■

Another way to show this (based on a comment by Nic):

Theorem 9: Let $f_n : A \rightarrow \mathbb{R}$ be a sequence of increasing functions defined on some set $A \subseteq \mathbb{R}$ such that the series $\sum_{n=1}^{\infty} f_n(x)$ converges pointwise for all $x \in A$. Then the function defined by the series is also increasing.

Proof: Let $(s_n(x))$ be the sequence of partial sums. Then each s_n is the finite sum of increasing functions, so it's increasing.

Since the series

$$s(x) = \sum_{n=1}^{\infty} f_n(x)$$

converges pointwise, we have $(s_n(x)) \rightarrow s(x)$ for all $x \in A$.

Let $x_1, x_2 \in A$ with $x_1 \leq x_2$. Then

$$s_n(x_1) \leq s_n(x_2)$$

for each $n \in \mathbb{N}$, so by the Order Limit Theorem,

$$s(x_1) \leq s(x_2),$$

i.e., s is increasing. ■